

Controllable Dynamic 3D Shape Generation via 3D Trajectories and Text

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Project Page: <https://cvlab-kaist.github.io/T2Mo>

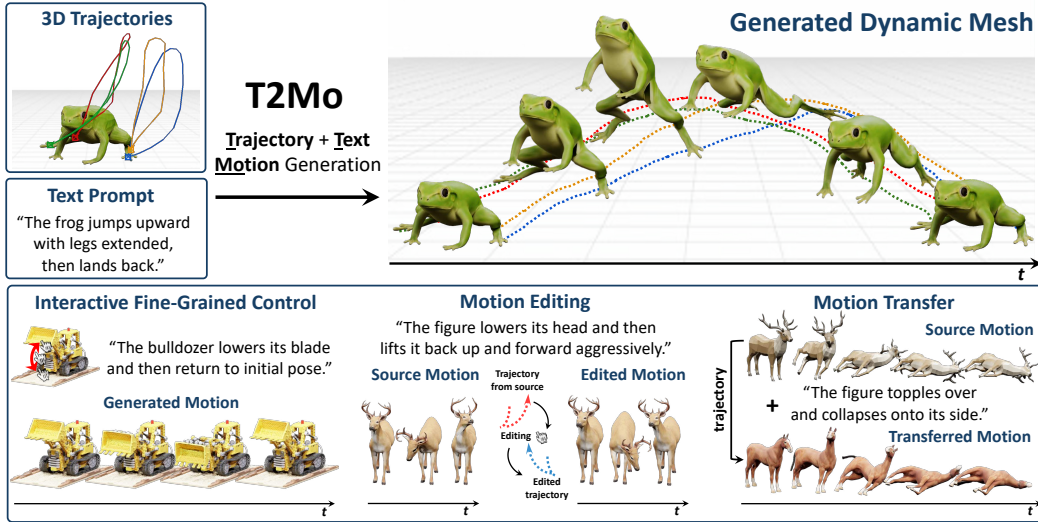


Figure 1: We present **T2Mo**, the first feed-forward framework for controllable dynamic 3D shape generation of general objects, conditioned on 3D trajectories and a text prompt. By combining the spatial guidance of 3D trajectories with the complementary semantic guidance of text, our framework faithfully generates specified motion while supporting a wide range of applications, including interactive fine-grained control, motion editing, and motion transfer.

Abstract

We present T2Mo, the first feed-forward framework for controllable dynamic 3D shape generation of general objects conditioned on 3D trajectories and text prompts. Existing text-driven methods struggle to specify precise motion due to the ambiguity of natural language. To address this, we introduce 3D trajectories as a direct spatial control signal and propose a shape-grounded trajectory embedding that maps arbitrary user-provided trajectories into geometry-aware condition tokens aligned with the input shape. Injecting these tokens into the generative backbone enables highly controllable motion generation while handling varying trajectory numbers and distributions. We conduct extensive comparisons against text-based baselines and trajectory-guided video generation workarounds. Quantitative and qualitative evaluations, along with user studies, show that our method produces motions that more faithfully follow the given prompts with higher expressiveness while preserving motion quality.

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1 Introduction

Dynamic 3D shape generation has become an important research problem because of its wide-ranging applications in AR/VR, gaming, filmmaking, and other domains that require temporally coherent 3D assets. Recent advances in feed-forward 3D generation models [12, 23–25, 27, 48, 55, 57] have made it possible to synthesize dynamic 3D shapes more efficiently than earlier optimization-based methods [2, 26, 29, 54] by extending these models in temporal dimension [19, 36, 52] or generating motion of given 3D object [4, 40, 46, 47], yet these models typically lack mechanisms for controlling the generated motion. Providing users with intuitive control over how objects move remains an open challenge, motivating the need for a controllable dynamic 3D shape generation framework.

While text prompts convey high-level intent, they often leave the precise motion unspecified. To address similar ambiguity in video synthesis [5, 10, 39, 41, 45, 53, 58] and human motion generation [11, 20, 34, 38], prior work introduces additional control signals such as bounding boxes, motion vectors, segmentation masks or kinematic constraints. These signals allow users to specify the desired motion in pixel space or on skeletons. However, they are not directly applicable to general dynamic 3D shapes represented as meshes, as there is no straightforward way to map 2D or domain-specific skeletal conditions onto the mesh geometry. Consequently, providing intuitive and expressive motion control for dynamic 3D generation remains an open problem.

We address this gap by representing user guidance directly on the mesh through *3D trajectories*. Trajectories provide a natural way to express motion control [10, 45, 58], encoding the path of a point across frames. They allow users to express spatial controls such as dragging a point along a curve or moving multiple points together. Because trajectories live in the same 3D domain as the generated asset, specifying motion in this way is both intuitive and interactive. Building on this representation, we introduce **T2Mo**, a feed-forward framework for controllable dynamic 3D generation of general objects. The framework conditions the generative model on an arbitrary set of user-provided trajectories together with a text prompt that provides global motion semantics. As a result, the interface naturally accommodates diverse types of control, such as interactive fine-grained control, motion editing and motion transfer, without being explicitly designed for each one.

A technical challenge arises from the fact that the number and spatial distribution of trajectories vary across instances and target motion. Depending on the input shape and intended motion, they may be highly sparse or concentrated in specific regions. Naively injecting such inconsistent trajectories into a generative model fails to adaptively localize the conditioning signal to the given shape. To address this, we design a *shape-grounded trajectory embedding* that maps trajectory set with arbitrary configuration into a dense, shape-aware token set that consistently covers the entire shape. This embedding strategy allows the model to consistently process arbitrarily distributed trajectories without losing geometric locality. By conditioning our model on this trajectory embedding and the text prompt, our model robustly generates the intended motion, achieving a high degree of controllability.

We evaluate T2Mo on a diverse set of objects and compare it against recent text-based baselines [40, 46] and trajectory-conditioned video generation-based approaches [4, 36, 58] through quantitative metrics including VBench [17], trajectory alignment, and motion magnitude [47], alongside user preference studies. The comparison indicates that incorporating trajectory guidance in generative pipeline improves both controllability and expressiveness by providing explicit motion cues, without compromising the overall quality of the generated motion. We further showcase qualitative results, demonstrating that our framework enables a diverse range of motion controls.

In summary, our contributions are as follows:

- We present **T2Mo**, the first feed-forward framework for controllable dynamic 3D shape generation of general objects, conditioned on *3D trajectories* and a *text prompt*.
- We introduce a **shape-grounded trajectory embedding** that maps arbitrarily configured input trajectories into a shape-aware token set defined over the entire object, allowing the model to consistently process diverse trajectory inputs while preserving geometric locality.
- Extensive experiments demonstrate that our framework enables a wide range of user-defined motion controls, improving both controllability and expressiveness over recent baselines without compromising the quality of the generated motion.

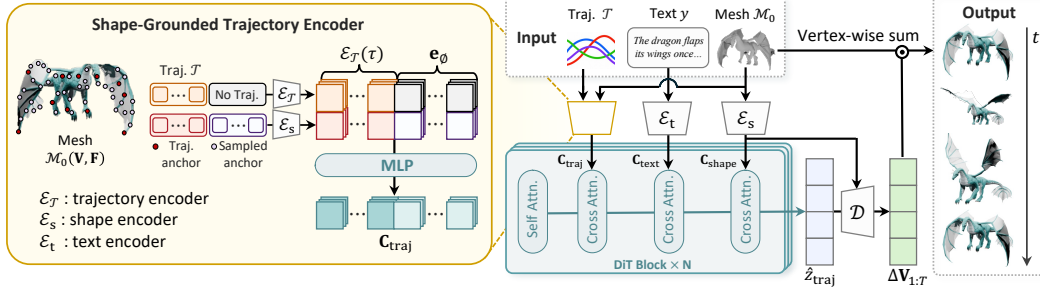


Figure 2: Overview of T2Mo. Given a source mesh $\mathcal{M}_0$, condition 3D trajectories $\mathcal{T}$, and a text prompt y , our model generates per-vertex displacements $\Delta \mathbf{V}_{1:T}$ via a DiT [30] backbone conditioned on shape $\mathbf{C}_{\text{shape}}$, text $\mathbf{C}_{\text{text}}$, and trajectory tokens $\mathbf{C}_{\text{traj}}$ through cross-attention, which yield the output mesh sequence $\{\mathcal{M}_i\}_{i=1}^T$ via the decoder $\mathcal{D}$. The trajectory condition is constructed by our *shape-grounded trajectory embedding* (yellow box): each input trajectory is bound to its source point on the mesh (red dots), and the remaining anchors are sampled by farthest point sampling (white dots). For each anchor, the shape feature from $\mathcal{E}_s$ is concatenated with either a trajectory feature from $\mathcal{E}_T$ or a learnable null embedding $\mathbf{e}_\emptyset$, and projected through an MLP to form the trajectory conditions $\mathbf{C}_{\text{traj}}$.

2 Related work

Dynamic 3D generation. Similar to static 3D generation, early work on dynamic 3D generation predominantly adopt optimization-based formulations [2, 26, 28, 29, 54]. Such methods commonly distill 2D generative prior through Score Distillation Sampling (SDS) [31] and its variants [42]. However, due to their limitations such as view inconsistency and the Janus problem [14], subsequent works shifted toward extending 2D generative models to multi-view settings [18, 32, 49, 56] for dynamic 3D reconstruction [9, 15, 16, 35, 37, 44]. Recently, as feed-forward 3D shape generation models [12, 23–25, 27, 48, 55, 57] have shown remarkable progress, there have been attempts to leverage these models to extend static 3D generation into dynamic generation [36, 52]. Alternatively, an emerging line of work orthogonally decomposes this task, independently handling static 3D asset creation and per-vertex trajectory synthesis [4, 19, 40, 46]. Building on this idea, recent text-guided dynamic 3D generation methods adopt the approach of animating predefined 3D meshes. Specifically, AnimateAnyMesh [46] and AnimateAnyMesh++ [47] utilize a rectified flow model with trajectory VAE to generate motion conditioned on a text and an initial shape. BiMotion [40] introduces a B-spline-based motion representation to handle motion of arbitrary frame lengths. Adopting this decoupling approach, we generate per-vertex displacements over time for a given input shape. This formulation is particularly suitable for controllable generation, as it allows users to directly provide guidance based on the underlying geometry. Although these advancements successfully generate dynamic 3D shapes from given contexts in feed-forward manner, they remain limited in providing the sufficient controllability required for practical applications. Motivated by this limitation, we propose a controllable generation approach for general objects through trajectory-based guidance.

Trajectory-guided motion control. Leveraging the spatiotemporal expressiveness of point trajectories, they have been widely adopted as a versatile representation for controllable generation across a range of tasks. Video generation, one of the most representative tasks, commonly employs sparse or dense trajectories on 2D pixel space as a spatial guidance of object or camera motion. Existing approaches integrate such trajectory guidance through feature-level modulation [5], ControlNet-style conditioning [10, 39], or specialized trajectory encoders [43, 45, 53, 58]. More recent extensions move beyond 2D image-plane trajectories by introducing 3D pose sequence [8] or trajectory with depth [39], mainly focusing on specifying the object-level global position trajectory. Similar principles have been applied to human motion synthesis [11, 20, 21, 34, 38, 50], utilizing trajectories as spatial constraints to guide global paths or specific actions, often conjugated with other condition modalities. However, these methods are largely limited to articulated human body motion, where the control mechanism is tightly coupled to a predefined skeletal structure. In the context of 4D generation, although TC4D [3] proposes trajectory-conditioned generation, trajectories are used solely to define the scene-level per-object position paths, rather than to model local object motion itself, leaving the local, intra-object motion uncontrolled. Unlike previous works, we propose a general-purpose framework that establishes 3D trajectories as a versatile interface for motion control.

3 Method

3.1 Overview

Our goal is to enable explicit control over dynamic 3D shape generation using two complementary signals: a *set of 3D trajectories* that specify point-wise motion on the mesh and a *text prompt* that conveys high-level motion semantics. However, in practice, user-provided trajectories are often sparse and unevenly distributed, and naively injecting them often fails to yield a well-localized conditioning signal on the shape. To handle such variability, we introduce **shape-grounded trajectory embedding**, which converts arbitrary sets of user trajectories into a fixed set of tokens anchored on the surface of the input shape. These tokens, together with text embeddings, condition a diffusion-based generative model built on DiT [30], which synthesizes per-vertex displacements for a given static mesh. This section first formalizes the task (Sec. 3.2), then describes the shape-grounded trajectory embedding and generative architecture (Sec. 3.3), and finally outlines training and inference procedures (Sec. 3.4).

3.2 Problem formulation

We formalize controllable dynamic 3D shape generation as generating a sequence of deformed meshes given a static source mesh and user-provided controls. Let $\mathcal{M}_0 = (\mathbf{V}_0, \mathbf{F}_0)$ be a source mesh with N vertices $\mathbf{V}_0 \in \mathbb{R}^{N \times 3}$ and M faces $\mathbf{F}_0 \in \mathbb{Z}^{M \times 3}$. The task is to generate per-vertex displacements $\Delta \mathbf{V}_{1:T} = \{\Delta \mathbf{V}_t\}_{t=1}^T$, where each $\Delta \mathbf{V}_t \in \mathbb{R}^{N \times 3}$, over T frames. Each output frame is reconstructed as $\mathcal{M}_t = (\mathbf{V}_0 + \Delta \mathbf{V}_t, \mathbf{F}_0)$. The user provides a text prompt y describing global motion semantics and a set of 3D trajectories $\mathcal{T} = \{\tau_k\}_{k=1}^K$ of arbitrary cardinality K . Each trajectory $\tau_k = (\mathbf{p}_{k,0}, \mathbf{p}_{k,1}, \dots)$ is a sequence of 3D waypoints, where $\mathbf{p}_{k,i} \in \mathbb{R}^3$ denotes the i -th waypoint.

Following prior work [40, 46, 47], we compress the static mesh and its motion into latent codes using a variational autoencoder (VAE). The VAE encoders map the source mesh and displacement sequence into a shape $\mathbf{z}_{\text{shape}}$ and a trajectory latent $\mathbf{z}_{\text{traj}}$ with two separate streams in the VAE encoder:

$$\mathbf{z}_{\text{shape}} = \mathcal{E}_{\text{shape}}(\mathcal{M}_0), \quad \mathbf{z}_{\text{traj}} = \mathcal{E}_{\text{traj}}(\Delta \mathbf{V}_{1:T}) \quad (1)$$

where $\mathcal{E}_{\text{shape}}$ and $\mathcal{E}_{\text{traj}}$ denote the shape and trajectory streams, respectively. At inference, a generative model $\mathcal{G}_\theta$ produces a trajectory latent $\hat{\mathbf{z}}_{\text{traj}}$ from Gaussian noise $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, conditioned on $\mathbf{z}_{\text{shape}}$ and the text and trajectory condition tokens $\mathbf{C}_{\text{text}}$ and $\mathbf{C}_{\text{traj}}$:

$$\hat{\mathbf{z}}_{\text{traj}} = \mathcal{G}_\theta(\epsilon; \mathbf{z}_{\text{shape}}, \mathbf{C}_{\text{text}}, \mathbf{C}_{\text{traj}}), \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad \Delta \mathbf{V}_{1:T} = \mathcal{D}(\mathbf{z}_{\text{shape}}, \hat{\mathbf{z}}_{\text{traj}}), \quad (2)$$

where $\mathcal{D}$ is the VAE decoder. For notational consistency, we hereafter define $\mathbf{C}_{\text{shape}} := \mathbf{z}_{\text{shape}}$.

3.3 T2Mo: Trajectory and text-guided dynamic 3D generation

Fig. 2 illustrates the overall architecture of **T2Mo**, including the generative model and the shape-grounded trajectory embedding.

Shape-grounded trajectory embedding. Simply treating each trajectory as an independent token neglects the geometry of the shape and leads to poorly localized conditioning (Fig. 5). Instead, we sample a fixed set of N_{anchor} anchor points $\{(\mathbf{s}_i, \mathbf{n}_i)\}_{i=1}^{N_{\text{anchor}}}$ on the surface, where each anchor consists of a position $\mathbf{s}_i \in \mathbb{R}^3$ and a surface normal $\mathbf{n}_i \in \mathbb{R}^3$. Specifically, we first place the source points $\{\mathbf{p}_{k,0}\}_{k=1}^{|\mathcal{T}|}$ of the given trajectory set $\mathcal{T}$ as trajectory anchors. We then fill the remaining anchor points with sampled anchors obtained by farthest point sampling seeded with the trajectory anchors. By construction, anchor $i \leq |\mathcal{T}|$ corresponds to trajectory τ_i . For each anchor i , we obtain a per-anchor embedding $\mathbf{e}_i$ from a trajectory embedding $\mathcal{E}_{\text{traj}}(\tau_i)$ when i is a trajectory anchor ($i \leq |\mathcal{T}|$), and from a learnable null embedding $\mathbf{e}_\emptyset$ otherwise. The conditioning token $\mathbf{c}_i$ for each anchor is then formed by concatenating the shape encoder’s output $\mathcal{E}_{\text{shape}}(\mathbf{s}_i, \mathbf{n}_i)$ with $\mathbf{e}_i$, followed by an MLP:

$$\mathbf{c}_i = \text{MLP}(\text{concat}(\mathcal{E}_{\text{shape}}(\mathbf{s}_i, \mathbf{n}_i), \mathbf{e}_i)), \quad \text{where} \quad \mathbf{e}_i = \begin{cases} \mathcal{E}_{\text{traj}}(\tau_i) & \text{if } i \leq |\mathcal{T}|, \\ \mathbf{e}_\emptyset & \text{otherwise.} \end{cases} \quad (3)$$

The resulting matrix $\mathbf{C}_{\text{traj}} \in \mathbb{R}^{N_{\text{anchor}} \times D}$ with feature dimension D provides dense, geometry-aware conditioning tokens that cover the entire shape and indicate which regions are guided by trajectories.

Generative architecture. Our generative model builds on DiT [30] and employs cross-attention to integrate the shape, trajectory and text conditions. For each DiT block, we compute keys and values

Table 1: Quantitative comparison with text-conditioned baselines. The *Cond.* column indicates the type of input conditions used by each method (text or trajectory). Our method outperforms the baselines [40, 46] across the user study, VBench [17], and AMD [47], while remaining competitive on motion smoothness (MS).

Method	Cond.		User Study $\uparrow$			VBench $\uparrow$				Motion Mag. $\uparrow$	
	Text	Traj	PA	MP	ME	OC	AQ	MS	DD	AMD	
AnimateAnyMesh [46]	✓	✗	2.170	2.667	2.560	0.168	0.539	0.992	0.585	0.043	
BiMotion [40]	✓	✗	2.657	3.040	2.813	0.170	0.536	0.991	0.329	0.046	
Ours	✓	✓	4.507	4.067	4.340	0.175	0.544	0.984	0.743	0.147	

from the shape tokens $\mathbf{C}_{\text{shape}}$ (via $\mathcal{E}_{\text{shape}}$), the trajectory tokens $\mathbf{C}_{\text{traj}}$ (via shape-grounded trajectory encoder), and text tokens $\mathbf{C}_{\text{text}}$ (via a CLIP text encoder [33]). We initialize the text and shape branches with weights from [40] and add the trajectory branch through a gated cross-attention [1] to smoothly incorporate trajectory guidance:

$$\begin{aligned} \mathbf{h} &\leftarrow \mathbf{h} + \tanh(\alpha) \cdot \text{Attn}(\mathbf{Q}_{\mathbf{h}}, \mathbf{K}_{\text{traj}}, \mathbf{V}_{\text{traj}}), \\ \mathbf{Q}_{\mathbf{h}} &= \mathbf{h} \mathbf{W}_q, \quad \mathbf{K}_{\text{traj}} = \mathbf{C}_{\text{traj}} \mathbf{W}_k, \quad \mathbf{V}_{\text{traj}} = \mathbf{C}_{\text{traj}} \mathbf{W}_v, \end{aligned} \quad (4)$$

where $\mathbf{h}$ denotes the hidden state and $\alpha \in \mathbb{R}$ is a learnable scalar gate initialized to zero, so the backbone initially ignores trajectory input.

3.4 Training and inference pipeline

Training. We train the model in two stages, inspired by the conditional diffusion training schemes in prior work [43, 53, 58]. Stage one conditions on trajectories assigned to all N_{anchor} anchors, teaching the model to use dense guidance. Stage two gradually sparsifies the guidance by randomly sampling n trajectories per sample, with n annealed from N_{anchor} down to a minimum value. We train the generative model using a rectified flow objective [7]. Given a noisy latent $\hat{\mathbf{z}}_s = (1 - s) \mathbf{z}_0 + s \mathbf{z}_1$ interpolated from random noise $\mathbf{z}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and a clean latent $\mathbf{z}_1$ with a scalar $s \sim \mathcal{U}(0, 1)$, the model predicts the velocity $\mathbf{v}_s = \mathbf{z}_1 - \mathbf{z}_0$ and minimizes:

$$\mathcal{L} = \mathbb{E}_{\mathbf{z}_0, \mathbf{z}_1, s, \mathbf{C}} \left[\left\| \mathbf{v}_{\theta}(\hat{\mathbf{z}}_s, s, \mathbf{C}) - (\mathbf{z}_1 - \mathbf{z}_0) \right\|_2^2 \right], \quad (5)$$

where $\mathbf{C} = \{\mathbf{C}_{\text{traj}}, \mathbf{C}_{\text{text}}, \mathbf{C}_{\text{shape}}\}$ denotes the conditioning tokens defined in Sec. 3.3.

Inference. At inference time, we solve the deterministic ODE associated with rectified flow. We apply classifier-free guidance [13] only to the text branch. Given the velocity fields predicted with only the trajectory tokens ($\mathbf{v}_{\theta}^{\text{traj}}$) and with both the text and trajectory tokens ($\mathbf{v}_{\theta}^{\text{both}}$), we compute the guided velocity as:

$$\mathbf{v}_{\theta}^{\text{cfg}} = \mathbf{v}_{\theta}^{\text{traj}} + \gamma \cdot (\mathbf{v}_{\theta}^{\text{both}} - \mathbf{v}_{\theta}^{\text{traj}}), \quad (6)$$

with a guidance scale γ . We do not guide the trajectory branch because the trajectory already explicitly provides its magnitude, and amplifying it tends to cause exaggerated motions. Users can specify arbitrary trajectories, which we discretize into waypoint sequences before passing them to the model.

4 Experiments

4.1 Experimental setup

Implementation details. We train a rectified flow-based generative model on the BiMO dataset, building on the pretrained VAE of [40]. Our model consists of 12 DiT blocks and is trained for 100K iterations at a learning rate of 5×10^{-5} , using the Adam optimizer [22] with weight decay 1×10^{-2} . We linearly warm up the learning rate over the first $1K$ iterations and then decay it to 1×10^{-5} via scheduled annealing. Training is performed on four NVIDIA RTX 3090 GPUs in 3 days. For each object, we randomly sample 4096 points from the source mesh and further select 512 of them via farthest point sampling to be encoded into latent tokens by the VAE encoder. For the shape-grounded trajectory embedding, we use $N_{\text{anchor}} = 64$ anchor points. The text condition is encoded with CLIP

Table 2: Quantitative comparison with cascaded video-based baselines. Our method achieves substantially lower trajectory error in both 2D pixel and 3D spaces, and outperforms the cascaded baselines across the user study, VBench [17], and AMD [47].

Method	Traj. Error ↓				User Study ↑			VBench ↑				Motion Mag. ↑
	ADE _{3D}	FDE _{3D}	ADE _{2D}	FDE _{2D}	PA	MP	ME	OC	AQ	MS	DD	AMD
Tora [58] + ActionMesh [36]	0.791	0.734	92.55	83.52	1.780	1.687	2.040	0.171	0.506	0.988	0.486	0.052
Tora [58] + Motion 3-to-4 [4]	0.605	0.675	69.19	77.64	1.940	1.800	1.903	0.166	0.522	0.991	0.486	0.042
Ours	0.089	0.071	12.52	8.52	4.577	4.177	4.437	0.175	0.544	0.984	0.743	0.147

ViT-L/14 [33], using a maximum token length of 77. For sampling, we use $\gamma = 3.0$ for classifier-free guidance [13]. To improve robustness to both coarse and noisy input trajectories, we perturb each conditioning trajectory with random Gaussian noise and additionally apply 1D Gaussian filtering and simplification-based smoothing. Further augmentation details are provided in Appendix A.2.

Baselines. We compare our model against text-conditioned baselines, AnimateAnyMesh [46] and BiMotion [40]. To further assess controllability, we additionally consider video-based baselines that incorporate trajectory conditioning indirectly by combining trajectory-guided video generation with video-based dynamic 3D generation. Specifically, we adopt Tora [58], a trajectory-conditioned video generator based on CogVideoX 5B [51] that accepts trajectories at arbitrary granularity, and pair it with two video-to-dynamic-3D generators, ActionMesh [36] and Motion 3-to-4 [4].

Evaluation protocol. Since BiMO [40] provides no evaluation set for dynamic meshes, we conduct quantitative evaluation on 30 animated meshes sourced from Objaverse [6] and Sketchfab, a scale consistent with prior work [40, 46], which uses 10 and 20 meshes, respectively. We assess the generated dynamic meshes along three perspectives: (i) *motion quality*, the visual fidelity of the generation; (ii) *motion expressiveness*, how dynamic the generated motion is; and (iii) *prompt alignment*, how faithfully the model follows the given text or trajectory condition. Following [18, 40, 46, 47], we evaluate these perspectives with VBench [17], reporting four sub-metrics: *aesthetic quality* (AQ) and *motion smoothness* (MS) for quality, *dynamic degree* (DD) for expressiveness, and *overall consistency* (OC) for prompt alignment. We additionally report *Average Motion Distance* (AMD) [47], which captures motion magnitude as the average inter-frame vertex displacement. We conduct a user study with 30 participants who rate the animations along three axes—*prompt alignment* (PA), *motion plausibility* (MP), and *motion expressiveness* (ME)—on a 5-point Likert scale ranging from 1 (*very poor*) to 5 (*excellent*) on 10 randomly chosen samples per baseline type, text-conditioned and video-generation-based baselines. For trajectory-conditioned baselines, we additionally report *Average Displacement Error* (ADE) and *Final Displacement Error* (FDE), which are the average ℓ_2 distance between each input trajectory and its corresponding generated vertices, taken over all frames and at the final frame, respectively. We measure these metrics in the 2D pixel space and 3D space for video-based baselines. All evaluations are run on 16-frame videos rendered from a fixed viewpoint; for VBench, we average the metrics over videos from two side viewpoints to mitigate viewpoint-dependent variance such as occlusion. Importantly, to reflect realistic settings where dense trajectories are unlikely to be provided, all quantitative evaluations use sparse trajectory sets of fewer than five trajectories per object; to rule out cherry-picking, these are sampled by a predefined rule-based procedure without per-instance selection or human intervention. For qualitative examples, only the trajectories shown in the figures are used. Additional details are provided in Appendix A.3.

4.2 Comparisons

Quantitative comparison with text-based baselines. Tab. 1 summarizes the quantitative comparison between our model and the text-based generation baselines [40, 46]. Our method outperforms these baselines across the user study, VBench [17], and AMD [47]. These results indicate that, by providing direct cues for the intended motion through the input trajectories, motion generated by our method follows the text prompt (PA, OC) more faithfully and produces higher quality motions (MP, AQ) and expressive motion (ME, DD, AMD) than the baselines. Although the baselines achieve slightly higher motion smoothness (MS) scores, MS rewards small frame-to-frame changes; together with the low ME in user study, AMD and dynamic degree (DD) scores of the baselines, this suggests that their MS scores stem from near-static or insufficiently dynamic motions.

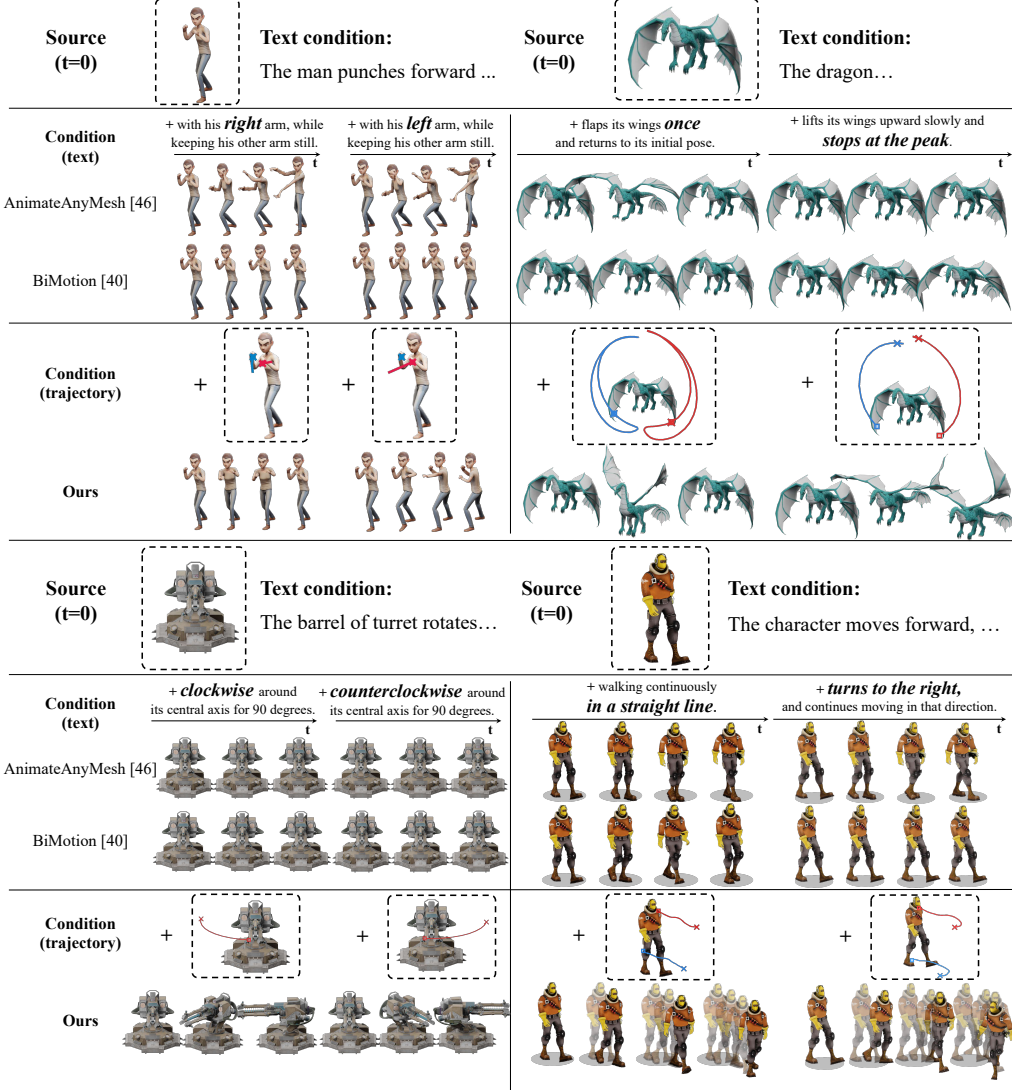


Figure 3: Qualitative comparison with text-based baselines. The baselines [40, 46] are conditioned on a pair of text prompts describing different specific motion details, while our model is guided by sparse trajectories representing those details. Our model uses only the visualized trajectories as inputs—at most two per example. By providing direct spatial cues, our method faithfully generates the intended motion, whereas the text-conditioned baselines fail to produce expressive motions aligned with the text and show little meaningful difference across the varying prompts. We provide video results in our supplementary materials.

Quantitative comparison with video-based baselines. Tab. 2 reports the comparison with baselines that we construct by combining controllable video generation [58] with video-based dynamic 3D generation [4, 36]. Across the VBench [17] and AMD [47], these baselines fail to generate motion of sufficient quality and expressiveness aligned with the trajectory and text prompt, similar to the trend observed in the text-based baselines. The user study results show an even larger gap in motion quality and expressiveness, indicating that the video-based workaround often fails to produce high-quality text-conditioned dynamic 3D shape generation. The trajectory error metrics (ADE and FDE), in both the 2D pixel and 3D spaces, are roughly an order of magnitude higher than our model, indicating that the control signal is not faithfully reflected. Compared to both types of baselines, our method produces more expressive motion while preserving quality and prompt alignment.

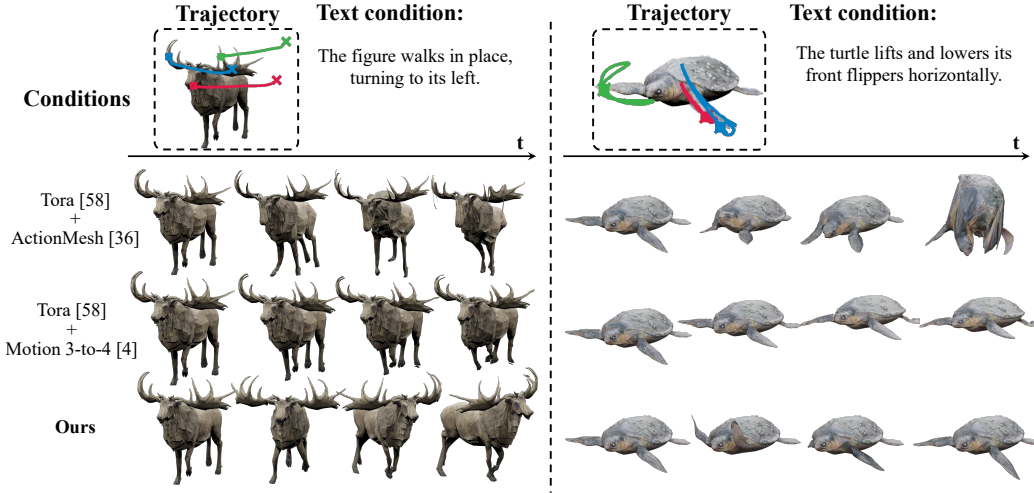


Figure 4: Qualitative comparison with video-based baselines. We compare our method against baselines that we construct by combining controllable video generation [58] with video-based dynamic 3D generation [4, 36]. These baselines take 2D trajectories projected onto the given camera view as input. The constructed pipelines fail to synthesize dynamic 3D shapes that faithfully follow the trajectory guidance and struggle to preserve the visual structure of the input shape during motion.

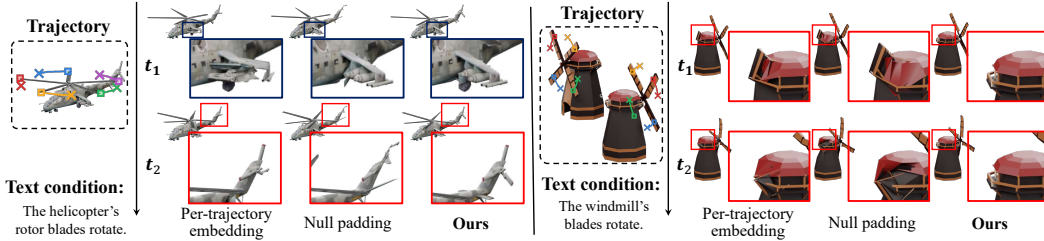


Figure 5: Qualitative comparison of trajectory embedding methods. We compare our shape-grounded trajectory embedding against two naive variants: per-trajectory embedding and null-padded per-trajectory embedding. With the naive variants, the trajectory conditions leak into unintended regions and produce noticeable artifacts, losing geometric locality. For visualization clarity, only the first 10% part of each trajectory is displayed.

Qualitative comparison with text-based baselines. We present qualitative results in Fig. 3 to compare the controllability, motion quality, and expressiveness of our generated motions against existing methods. To demonstrate the superior controllability of our trajectory-based approach over text-conditioned baselines [40, 46], we task the models with generating specific motions: the text-based baselines are provided with detailed text descriptions, while our model is guided by sparse trajectories. As shown in the Fig. 3, while the text-conditioned baselines struggle to realize finely specified motions from text prompts alone, our method successfully produces the intended dynamics by leveraging direct spatial guidance. This includes detail specification (top left), precise temporal control (top right), directional alignment (bottom left), and global trajectory tracking (bottom right).

Qualitative comparison with video-based baselines. We further compare with the video-based baselines [4, 36, 58] in Fig. 4. Our method generates motion that reflects both the trajectory and text conditions, whereas the combined baselines neither preserve the visual quality of the input shape nor faithfully follow the given trajectory prompt.

Table 3: Ablation study on the trajectory embedding. Top: comparison of trajectory embedding variants. **Bottom:** effect of the number of anchor points N_{anchor} .

Method	Traj. Error ↓		VBench ↑			
	ADE _{3D}	FDE _{3D}	OC	AQ	MS	DD
<i>Trajectory embedding strategy</i>						
Per-traj. emb.	0.159	0.124	0.173	0.541	0.985	0.771
Null padding	0.101	0.080	0.174	0.540	0.984	0.743
Ours ($N_{\text{anchor}} = 64$)	0.089	0.071	0.175	0.544	0.984	0.743
<i>Number of anchor points for training</i>						
Ours ($N_{\text{anchor}} = 32$)	0.091	0.070	0.176	0.548	0.984	0.714
Ours ($N_{\text{anchor}} = 64$)	0.089	0.071	0.175	0.544	0.984	0.743
Ours ($N_{\text{anchor}} = 128$)	0.092	0.074	0.175	0.544	0.984	0.714
Ours ($N_{\text{anchor}} = 256$)	0.090	0.071	0.176	0.543	0.984	0.743

Table 4: Stress test on conditioning trajectories. Downsampled: uniformly subsampled to N waypoints. The first and last waypoints are preserved. **Perturbed:** Gaussian noise with standard deviation σ is added to each waypoint in the normalized $[-0.9, 0.9]$ coordinate space.

Trajectory Type	Traj. Error ↓		VBench ↑			
	ADE _{3D}	FDE _{3D}	OC	AQ	MS	DD
Original	0.089	0.071	0.175	0.544	0.984	0.743
Downsampled ($N = 8$)	0.093	0.080	0.174	0.544	0.985	0.729
Downsampled ($N = 4$)	0.121	0.090	0.174	0.543	0.986	0.700
Perturbed ($\sigma = 0.05$)	0.093	0.075	0.175	0.544	0.984	0.743
Perturbed ($\sigma = 0.1$)	0.100	0.085	0.174	0.542	0.984	0.743

4.3 Ablations

Trajectory embedding method. To validate our shape-grounded trajectory embedding method, we compare it against two variants: (i) *per-trajectory embedding*, which directly tokenizes each input trajectory and produces a variable number of condition tokens; and (ii) *per-trajectory embedding padded with null padding*, which pads the null tokens to a fixed size to remove for the effect of token sparsity. As shown in the top section of Tab. 3, our shape-grounded trajectory embedding achieves the lowest trajectory alignment error, consistent with its design of providing spatially consistent conditioning. It also obtains slightly higher VBench scores in aesthetic quality (AQ) and overall consistency (OC). The per-trajectory variant without padding obtain a higher dynamic degree (DD); however, this comes alongside substantially worse trajectory alignment error, and the qualitative comparisons in Fig. 5 show that the additional motion often appears in regions not specified by the input trajectories. This suggests that, when a spatially inconsistent and unevenly distributed trajectory set is naively injected, the model struggles to provide localized control over the intended regions. While this phenomenon is rarely observed in standard video generation, it emerges as a critical challenge in dynamic 3D generation. We suggest that this is because trajectory conditioning in video generation operates within a fixed 2D pixel space where the background inherently provides dense, global guidance. In our 3D setting, however, the spatial domain is highly shape-dependent, and the sparse input trajectories alone cannot provide sufficient global coverage across the entire surface.

Effect of the number of anchor points. We further conduct an ablation study on the number of anchor points to examine its effect on performance. We report the results of varying N_{anchor} from 32 to 256 in the second section of Tab. 3. The performance remains stable across this range, indicating that our shape-grounded trajectory embedding is robust to the choice of N_{anchor} . We set $N_{\text{anchor}} = 64$ in all experiments as it achieves strong trajectory alignment with comparable VBench scores.

Trajectory quality stress test. In real-world settings, user-provided trajectories may be noisy or coarse compared to the ground-truth trajectories used during training. To evaluate the robustness of our model against such imperfect inputs, we test two perturbations: (i) *downsampled* trajectories, obtained by uniformly sampling N waypoints ($N < 16$) while preserving the first and last waypoints; and (ii) *perturbed* trajectories, obtained by adding Gaussian noise with a standard deviation of σ to each waypoint (within a $[-0.9, 0.9]$ normalized coordinate space). In both cases, trajectory errors are computed against the original ground-truth trajectories. As reported in Tab. 4, while trajectory errors (ADE, FDE) inevitably increase, this degradation is marginal, and the video-based VBench metrics remain largely preserved. This indicates that the model robustly synthesizes the intended motion even when conditioned on coarse or moderately noisy trajectories.

5 Conclusion

We presented T2Mo, a feed-forward framework for controllable dynamic 3D shape generation conditioned on 3D trajectories and a text prompt. The key component is a shape-grounded trajectory embedding that maps an arbitrarily configured trajectory set into a dense, geometrically aware token set covering the entire shape, which enables stable conditioning under any input configuration. Across both quantitative and qualitative evaluations, our framework produces motion that more faithfully

reflects the given prompts than the text-conditioned and video-based baselines, while supporting various applications and user-control signals such as sparse-trajectory dragging, motion editing, and motion transfer. We believe this approach makes a practical step toward enabling intuitive and expressive control in dynamic 3D shape generation.

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A Technical appendices and supplementary material

A.1 Implementation details

Training details. Inspired by previous work on trajectory-guided generation [43, 53, 58], we adopt a two-stage training strategy. In the first stage, trajectories are sampled at all N_{anchor} anchor points of the shape-grounded trajectory embedding, so that the model first learns to exploit the trajectory condition in a stable manner. In the second stage, the number of sampled trajectories is drawn uniformly at random between n_{min} and n_{max} , where n_{max} is progressively annealed from N_{anchor} down to n_{min} over the course of training. To encourage the model to learn meaningful trajectory guidance, a fixed proportion ρ of the conditioning trajectories at each step is sampled exclusively from vertices with high motion magnitude, while the remaining trajectories are sampled uniformly at random across all vertices. Specifically, we draw these high-motion trajectories from the top 30% of vertices ranked by motion magnitude and use $\rho = 0.7$.

Architecture details. We build on the pretrained B-spline-based VAE of [40] and reuse the same VAE to embed the trajectory conditions for efficient training. The weights of the VAE encoder, including its two streams $\mathcal{E}_{\text{shape}}$ and $\mathcal{E}_{\text{traj}}$, are not updated during trajectory conditioning. We follow the original implementation for all transformations involved in the B-spline-based representation. Specifically, the cross-attention weight sharing in the original implementation is retained when computing the latent embeddings $\mathbf{z}_{\text{shape}}$ and $\mathbf{z}_{\text{traj}}$, but is omitted when encoding the trajectory condition to reduce the influence of the spatial configuration of the conditioning trajectories and the dilution introduced by the null embeddings. For brevity, we omit the cross-attention weight sharing in our notation. AdaLN [30] conditioned on diffusion timestep is applied independently to the noisy latent $\hat{\mathbf{z}}_{\text{traj}}$ and each conditioning token.

Anchored farthest point sampling. Our shape-grounded trajectory embedding requires a fixed set of N_{anchor} anchor points that (i) include the starting positions of all user-provided trajectories so that each trajectory is grounded at its intended location on the shape, and (ii) cover the rest of the shape to provide geometric context for unconditioned regions. We achieve both with an anchored variant of farthest point sampling. Given a trajectory set $\mathcal{T} = \{\tau_k\}_{k=1}^{|\mathcal{T}|}$, we first take the starting waypoints $\{\mathbf{p}_{k,0}\}_{k=1}^{|\mathcal{T}|}$ as the trajectory anchors and include them as the first $|\mathcal{T}|$ entries of the anchor set. The remaining $N_{\text{anchor}} - |\mathcal{T}|$ anchors are then selected from the mesh vertices $\mathbf{V}_0$ by farthest point sampling seeded with the trajectory anchors, so that newly added anchors are progressively chosen to maximize the minimum distance to the existing anchor set. During training, each trajectory anchor coincides with a mesh vertex; at inference, when a user-provided trajectory does not start exactly on a vertex, its starting position is snapped to the nearest point on the mesh surface before serving as an anchor. This procedure guarantees that the trajectory anchors are preserved exactly while the remaining anchors evenly cover the shape.

A.2 Data augmentation

To improve robustness and encourage the model to actively utilize trajectory information, we apply two categories of data augmentations. All augmentations described below are applied only when trajectories are provided as conditions, since they perturb the data distribution that text alone could otherwise capture and may degrade text-conditioned generation. First, we apply global geometric and temporal augmentations. We inject random noise into the input shape (positions and normals) as well as the target and conditioning trajectories, alongside random horizontal rotation and flipping. Additionally, we apply temporal cropping that retains at least 70% of the full sequence length to preserve the original motion semantics. These transformations not only enhance overall robustness but also encourage the model to resolve spatial ambiguities—which cannot be inferred from text alone—by relying on the trajectory guidance. Second, to ensure the model is robust to the coarse and noisy trajectories typically provided by users, we apply targeted augmentations exclusively to the conditioning trajectories. This involves injecting Gaussian noise, followed by two smoothing techniques: a simplification step that subsamples a random subset of intermediate waypoints while retaining the endpoints, and 1D Gaussian-kernel smoothing applied along the temporal axis with a randomly chosen kernel width, which suppresses high-frequency jitter and yields smoother trajectories closer to those a user would draw.

A.3 Evaluation details

Evaluation metrics. We represent each generated motion as a sequence of T deformed meshes. Let $\mathbf{V}_t \in \mathbb{R}^{N \times 3}$ denote the vertex positions of the generated mesh at frame $t \in \{1, \dots, T\}$, where N is the number of vertices and $(\mathbf{V}_t)_i$ is the 3D position of vertex i .

For evaluation, we are given a mesh with motion together with K target trajectories used as conditioning. Each trajectory has its source point on the mesh surface, where the source point is included with a mesh vertex, i.e., $\mathbf{p}_{k,0} \in \mathbf{V}_0$. We denote the corresponding vertex index by i_k , so that $(\mathbf{V}_0)_{i_k} = \mathbf{p}_{k,0}$. The generated vertex $(\mathbf{V}_t)_{i_k}$ is then compared with the target waypoint $\mathbf{p}_{k,t}$ at each frame.

Average Motion Distance (AMD) measures the average inter-frame displacement over all vertices:

$$\text{AMD} = \frac{1}{T-1} \sum_{t=1}^{T-1} \frac{1}{N} \sum_{i=1}^N \|(\mathbf{V}_{t+1})_i - (\mathbf{V}_t)_i\|_2. \quad (7)$$

AMD measures the overall amount of generated motion, not trajectory-tracking accuracy. It is computed in 3D world coordinates and reported in the same units as the mesh.

Average Displacement Error (ADE) and *Final Displacement Error (FDE)* measure trajectory-tracking accuracy for the K controlled vertices. For the k -th trajectory, we define

$$\text{ADE}_k = \frac{1}{T} \sum_{t=1}^T \|(\mathbf{V}_t)_{i_k} - \mathbf{p}_{k,t}\|_2, \quad \text{FDE}_k = \|(\mathbf{V}_T)_{i_k} - \mathbf{p}_{k,T}\|_2. \quad (8)$$

The final ADE and FDE are obtained by averaging over all input trajectories:

$$\text{ADE} = \frac{1}{K} \sum_{k=1}^K \text{ADE}_k, \quad \text{FDE} = \frac{1}{K} \sum_{k=1}^K \text{FDE}_k. \quad (9)$$

For the ADE_{3D} and FDE_{3D} , the above distances are computed directly in world coordinates. For the 2D variant, both the generated vertex and the target waypoint are projected to the image plane using the evaluation camera projection $\pi : \mathbb{R}^3 \rightarrow \mathbb{R}^2$:

$$\text{ADE}_{2D,k} = \frac{1}{T} \sum_{t=1}^T \|\pi((\mathbf{V}_t)_{i_k}) - \pi(\mathbf{p}_{k,t})\|_2, \quad \text{FDE}_{2D,k} = \|\pi((\mathbf{V}_T)_{i_k}) - \pi(\mathbf{p}_{k,T})\|_2. \quad (10)$$

The 2D errors are reported in pixels and are averaged over the fixed evaluation viewpoints.

Condition trajectory selection. For quantitative evaluation, we obtain the input trajectories for our model automatically, without human intervention, by sampling from the per-vertex ground-truth trajectories using a two-stage predefined rule-based procedure. This process is performed to avoid bias introduced by subjective trajectory selection. In the first stage, we score each vertex by its cumulative path length over the sequence and draw $N_{\text{cand}} = 64$ candidates from the top-30% high-motion vertices, with sampling probabilities biased toward higher-motion vertices by a power law with exponent $\gamma = 2$. In the second stage, the candidates are clustered using K-Means on a feature composed of each vertex’s source position and its per-frame displacements. The number of clusters $K \in \{3, 4, 5\}$ is selected by the silhouette score, and the trajectory closest to each cluster centroid is retained as the representative. The resulting set $\mathcal{R}$, with $|\mathcal{R}| = K \leq 5$, is used as the trajectory input for quantitative evaluation. For qualitative figures, we use the same automatic procedure unless otherwise specified. When specified trajectories are used, we explicitly visualize the conditioning trajectories in the figure. The average number of conditioning trajectories is 4.01 in quantitative evaluation, and no hyperparameter search was performed to optimize the evaluation results.

User study protocol. Since subtle aspects of motion are difficult to evaluate with automatic pipelines, we conduct an anonymous user study to obtain human perceptual judgments. The study involves 30 participants, each rating the generated animations along three axes: *Prompt Alignment* (PA), *Motion Plausibility* (MP), and *Motion Expressiveness* (ME). Ratings are reported on a 5-point Likert scale, where 1 denotes *very poor* and 5 denotes *excellent*. We evaluate against two groups of baselines: text-conditioned baselines [40, 46] that take only text as input, and cascaded video-based baselines [4, 36, 58] that take both text and trajectories. For each group, we independently sample 10

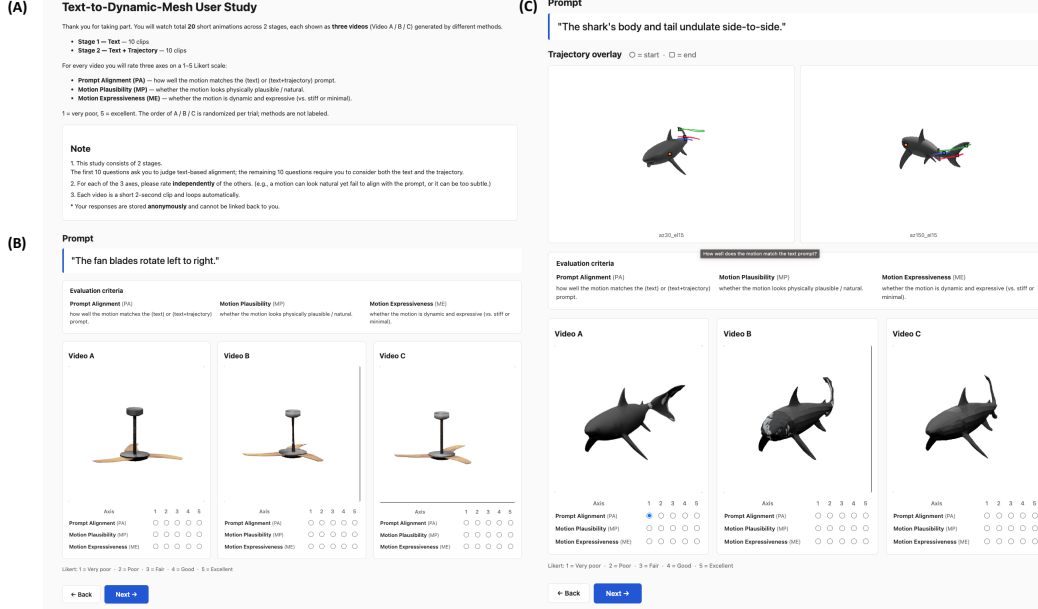


Figure 6: User study interface. (A) Initial instruction page describing the rating task and evaluation criteria. (B) Rating page for text-conditioned baselines, where participants rate three videos generated by different methods given a text prompt. (C) Rating page for cascaded video-based baselines, where participants additionally see the input trajectory overlay alongside the text prompt. Videos from compared methods are presented side-by-side in a randomly shuffled order.

Table 5: User study results. Each cell shows the mean $\pm$ standard deviation over all participants and examples. *Left:* text-based baselines [40, 46]. *Right:* video-based baselines. [4, 36, 58]

Methods	PA $\uparrow$	MP $\uparrow$	ME $\uparrow$
AnimateAnyMesh [46]	2.170 $\pm$ 1.260	2.667 $\pm$ 1.452	2.560 $\pm$ 1.243
BiMotion [40]	2.657 $\pm$ 1.525	3.040 $\pm$ 1.536	2.813 $\pm$ 1.499
Ours	4.507$\pm$0.782	4.067$\pm$1.055	4.340$\pm$0.880

Methods	PA $\uparrow$	MP $\uparrow$	ME $\uparrow$
ActionMesh [36]	1.780 $\pm$ 1.004	1.687 $\pm$ 1.042	2.040 $\pm$ 1.053
Motion 3-to-4 [4]	1.940 $\pm$ 1.071	1.800 $\pm$ 1.025	1.903 $\pm$ 0.944
Ours	4.577$\pm$0.657	4.177$\pm$0.921	4.437$\pm$0.775

(a) Text-based baselines

(b) Video-based baselines

objects at random from the evaluation set. For the video-based baselines, prompt alignment (PA) is assessed against both the trajectory condition and the text prompt, since these models receive both conditions. In total, each participant answers $(10 + 10) \times 3 \times 3 = 180$ questions, covering 20 objects, three baselines, and three rating axes. As shown in Fig. 6, participants are first presented with the rating instructions, after which they rate each object using a video and the corresponding prompt. For each object, the videos from all compared methods are presented side-by-side in a randomly shuffled order so that participants are blind to the source of each video, and each video is automatically looped. A consolidated summary of the user study results, including standard deviations, is reported in Tab. 5.

Video generation procedure. To evaluate the cascaded video-based baselines which combines trajectory-conditioned video generation model with video-to-dynamic mesh generation model, we first generate trajectory-conditioned videos using Tora [58]. We choose Tora as the video-generation baseline because it is a publicly available trajectory-conditioned video generator that supports arbitrary point-level 2D trajectory control, rather than relying on object-level controls. Since Tora operates in the image plane, we project the input 3D trajectories into 2D pixel coordinates using the same camera viewpoint as the rendered videos used for VBench [17]. To avoid degraded video quality caused by occluded trajectories, we discard any trajectory whose source vertex is occluded in the first frame; occlusion is determined by comparing the projected depth of each vertex with the depth value at the corresponding pixel. The resulting videos, together with the ground-truth source mesh, are then passed to the video-to-dynamic-3D mesh generation pipelines [4, 36], which produce the final dynamic mesh sequence used for evaluation. All baselines are run with their publicly available models and default configurations.

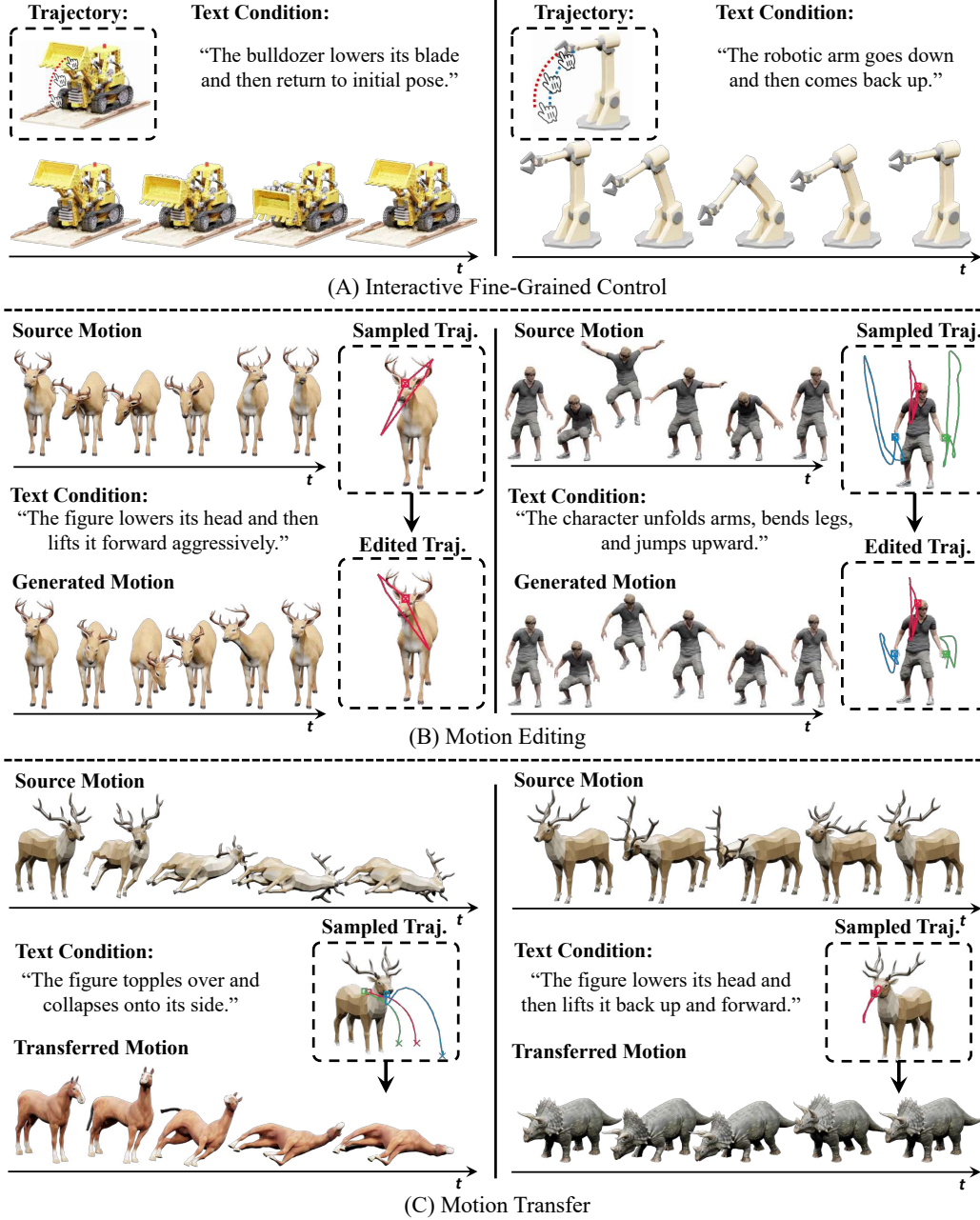


Figure 7: Application showcase. Our framework supports a diverse range of motion controls through joint trajectory and text conditioning. *Top*: fine-grained motion control with user-specified trajectories. *Middle*: motion editing by modifying a few waypoints of representative trajectories. *Bottom*: motion transfer from a source motion to a target mesh via sampled representative trajectories. Our model uses only the visualized trajectories as input.

A.4 Applications

Trajectories provide a flexible medium for expressing various forms of motion control. By accepting trajectory sets of arbitrary configuration, our model exploits this flexibility to enable diverse applications, including interactive fine-grained control, motion editing, and motion transfer.

Interactive fine-grained control. It is challenging to describe the precise motion which a user imagines, and as shown in Fig. 3, models conditioned solely on text often struggle to faithfully reflect detailed prompts. In contrast, our model accepts explicit spatial trajectories, allowing users to specify motion attributes such as direction and magnitude. As shown in the first row of Fig. 7, by reflecting user-provided trajectories together with the global semantics of the text prompt, our model enables fine-grained motion generation in an interactive manner, specifying the direction and magnitude.

Motion editing. Refining a previously generated or pre-existing motion is a time-consuming and laborious task: without rigging, it requires directly adjusting the movements of many individual vertices, and even with rigging, multiple skeletal motions must be coherently edited manually. Since our model can generate motions consistent with a sparse set of input trajectories, editing reduces to modifying a few waypoints of a small number of representative trajectories in the region of interest. Specifically, given a set of representative trajectories $\mathcal{T}_{\text{rep}} = \{\tau_k\}_{k=1}^{|\mathcal{T}_{\text{rep}}|}$ sampled from the source motion, the user edits a subset of waypoints indexed by $\mathcal{I} \subseteq \{1, \dots, |\mathcal{T}_{\text{rep}}|\} \times \{1, \dots, T\}$. For each $(k, t) \in \mathcal{I}$, the original waypoint $\mathbf{p}_{k,t}$ is replaced with an edited waypoint $\mathbf{p}_{k,t}^*$, while all other waypoints are kept unchanged. This yields an edited trajectory set $\mathcal{T}_{\text{rep}}^*$. We then pass $\mathcal{T}_{\text{rep}}^*$, the text prompt, and a target frame of the source motion as the source mesh to our model, which generates the edited motion. We show examples in the second row of Fig. 7.

Motion transfer. Transferring a source motion to another mesh in a spatially consistent manner is difficult to achieve with text alone, as text cannot impose precise spatial constraints, and is also unstable with video-based approaches due to topology mismatch. In contrast, 3D trajectories provide a topology-agnostic motion representation, since they specify point-wise paths in space rather than depending on the mesh connectivity or the reference topology. By sampling a few representative trajectories from a source object and attaching them to the corresponding vertices of a target mesh, our model performs motion transfer without additional supervision. Specifically, given a source motion defined on a mesh $\mathcal{M}_{\text{src}} = (\mathbf{V}_{\text{src}}, \mathbf{F}_{\text{src}})$ with per-vertex trajectories $\mathcal{T}_{\text{src}} = \{\tau_k\}_{k=1}^{|\mathbf{V}_{\text{src}}|}$, we sample a sparse subset $\mathbf{V}_{\text{src}}^* \subset \mathbf{V}_{\text{src}}$ of representative vertices and their associated trajectories $\mathcal{T}_{\text{src}}^* \subset \mathcal{T}_{\text{src}}$. We attach $\mathcal{T}_{\text{src}}^*$ to the corresponding vertices $\mathbf{V}_{\text{tgt}}^*$ of the target mesh $\mathcal{M}_{\text{tgt}} = (\mathbf{V}_{\text{tgt}}, \mathbf{F}_{\text{tgt}})$, and feed the text prompt, the attached trajectories, and $\mathcal{M}_{\text{tgt}}$ to our model to transfer the source motion onto the target mesh. We showcase examples in the last row of Fig. 7, where only three representative trajectories are sampled from the source motion.

A.5 Social impact and limitations

Social impact. This work aims to make dynamic 3D content creation more accessible by enabling users to control mesh motion through sparse 3D trajectories and natural language. Such controllability can reduce the amount of manual labor required for animation, particularly for users without expertise in rigging, keyframing, or skeletal motion editing, thereby supporting more efficient workflows in digital content creation, AR/VR, gaming, filmmaking, education, and simulation. At the same time, controllable dynamic 3D generation may introduce risks. Generated animated assets could be misused to create deceptive, misleading, or harmful visual content, especially when combined with realistic rendering. The ability to edit or transfer motion may also raise concerns about unauthorized reuse of copyrighted assets, character designs, or motion data. In addition, if training data contain biased or inappropriate content, generated motions may inherit or amplify such biases. These risks highlight the importance of responsible data curation, compliance with dataset licenses, clear usage guidelines, and provenance mechanisms such as watermarking or metadata for generated assets. We encourage release practices that include documentation of training data sources, intended use cases, and known limitations.

Limitations. Although T2Mo offers a high degree of controllability via 3D trajectories and text, our formulation predicts per-vertex displacements on a fixed source mesh and therefore assumes a constant topology across frames. Consequently, motions involving topological changes, such as splitting, merging, or the emergence of new components, lie outside the scope of our model. A future direction is to integrate our shape-grounded trajectory embedding with per-frame generation methods, enabling trajectory-controllable dynamic 3D generation under varying topology, thereby extending our framework beyond motion synthesis to broader dynamic scene generation where the underlying geometry itself evolves over time.